

4 Microbial activity chelation and root stimulation

Biology can assist nutrient cycling and root growth, but performance depends on habitat, organism, crop, and product quality.

What is well established

- **Organic residues feed soil organisms.** Microbes decompose materials and release or temporarily hold nutrients as part of normal cycling.
- **Roots and microbes exchange compounds.** Roots release sugars and organic compounds; nearby organisms use them and alter the chemistry around roots.
- **Some microbes release organic acids.** These acids can help dissolve mineral-bound phosphorus or other nutrients in small zones near roots.
- **Some microbes make chelating compounds.** These compounds bind metals such as iron and can change their mobility and availability.
- **Healthy habitat matters.** Moisture, oxygen, temperature, pH, salinity, carbon supply, and living roots influence microbial activity.

PGPR in plain language

Plant growth-promoting rhizobacteria or PGPR are beneficial bacteria that live near roots. Depending on the strain and setting, they may influence root branching, nutrient availability, stress response, or competition with pathogens.

Claim	Evidence label	Practical interpretation
PGPR can stimulate root or shoot growth in trials.	Established mechanism; context dependent result	Plausible and documented, but the effective strain and crop combination matters.
A commercial inoculant will work in every avocado grove.	Product dependent and unproven as a general rule	Require identity, viability, storage, rate, field replication, and a clear comparison.
Organic acids or chelates can improve micronutrient availability.	Established chemistry; context dependent field effect	The compound, pH, metal, dose, persistence, and root condition determine the result.
More microbial respiration is always better.	Incomplete	High activity can reflect cycling, but interpretation also depends on moisture, disturbance, carbon inputs, and nutrient balance.

Minimum screen for biological or chelate products

- **Define the target and material.** Name the intended outcome; record microbial strain count and viability or the chelate ligand and nutrient form.
- **Check the habitat first.** A product cannot reliably overcome waterlogging severe salinity compaction or a failed irrigation system.
- **Use a fair comparison.** Include a similar untreated area and measure the same outcomes before and after treatment.
- **Judge field value.** Track roots new growth yield quality and relevant tests rather than relying only on a claim.